

# Multimodal Deep Learning

## Solutions to Exercise Sheet 6

Date: 24th of June, 2025

### Exercise 1: Information-Theory Basics

Symbols  $w$  are drawn from a finite alphabet  $W$ . Let  $p(w)$  be the *true* distribution and  $q_\theta(w)$  a model family parametrized by  $\theta$ .

- (a) Which information-theoretic quantity should be *minimized* to make  $q_\theta$  approach the true distribution  $p$ ?

*Hint:* Two equivalent answers are valid under the support assumption.

- (b) Explain why the quantity from (a) is a sensible objective and show the explicit connection

$$H(p, q_\theta) = -\mathbb{E}_p[\log q_\theta(w)].$$

Why does this imply that *maximizing* the (average) log-likelihood is equivalent to *minimizing* cross-entropy?

- (c) Show that

$$H(p, q_\theta) = H(p) + D_{\text{KL}}(p \| q_\theta).$$

Conclude that, because  $H(p)$  is constant in  $\theta$ , minimizing cross-entropy is equivalent to minimizing the forward KL divergence.

- (d) Analyse what goes wrong in the argument of (c) if  $q_\theta(w) = 0$  for some  $w$  with  $p(w) > 0$ .  
*Hint:* Discuss both the mathematical value and its practical implications for optimization.

**Empirical setting** Draw  $n$  i.i.d. samples  $w_{1:n} \sim p$  and let  $\hat{p}_n$  be their empirical distribution. Show that

$$\text{NLL}(\theta) = -\frac{1}{n} \sum_{i=1}^n \log q_\theta(w_i) = H(\hat{p}_n, q_\theta),$$

and hence that minimizing NLL is the same as minimizing  $D_{\text{KL}}(\hat{p}_n \| q_\theta)$ .

*Remark:* Explain why driving this empirical KL to zero need not guarantee good *generalization* to unseen data.

- (e) **Mutual information forms**

For  $(X, Y) \sim p_{XY}$ , starting from  $I(X; Y) = D_{\text{KL}}(p_{XY} \| p_X p_Y)$ :

1. Derive the equivalent entropy form  $I(X; Y) = H(X) + H(Y) - H(X, Y)$ .
2. Give a one-sentence interpretation of the case  $I(X; Y) = 0$ .

- (f) **Numerical sanity check** Let  $W = \{0, 1\}$  with  $p(1) = 0.9$ ,  $p(0) = 0.1$  and model  $q_\theta(1) = 0.8$ ,  $q_\theta(0) = 0.2$ . Using base-2 logarithms compute:

1.  $H(p)$ ,
2.  $H(p, q_\theta)$ ,
3.  $D_{\text{KL}}(p \| q_\theta)$ ,

and verify numerically that  $H(p, q_\theta) = H(p) + D_{\text{KL}}(p \| q_\theta)$ .

**Solution**

(a) To make  $q_\theta$  approach  $p$ , one should minimize either:

1. The Kullback-Leibler (KL) divergence:  $D_{\text{KL}}(p\|q_\theta) = \sum_w p(w) \log \frac{p(w)}{q_\theta(w)}$ .
2. The cross-entropy:  $H(p, q_\theta) = -\sum_w p(w) \log q_\theta(w)$ .

These are equivalent objectives under the support assumption  $\{w : p(w) > 0\} \subseteq \{w : q_\theta(w) > 0\}$ .

(b) The KL divergence  $D_{\text{KL}}(p\|q_\theta) \geq 0$ , with equality if and only if  $p = q_\theta$ . Minimizing it thus drives  $q_\theta$  towards  $p$ . The cross-entropy is  $H(p, q_\theta) = -\sum_w p(w) \log q_\theta(w)$ . By definition of expectation,  $\mathbb{E}_p[f(w)] = \sum_w p(w)f(w)$ . Letting  $f(w) = \log q_\theta(w)$ , we get

$$H(p, q_\theta) = -\mathbb{E}_p[\log q_\theta(w)].$$

Maximizing the average log-likelihood  $\frac{1}{N} \sum_i \log q_\theta(w_i)$  (which for large  $N$  approaches  $\mathbb{E}_p[\log q_\theta(w)]$ ) is equivalent to minimizing its negative,  $-\mathbb{E}_p[\log q_\theta(w)]$ , which is  $H(p, q_\theta)$ .

(c)

$$\begin{aligned} H(p, q_\theta) &= -\sum_w p(w) \log q_\theta(w) \\ &= -\sum_w p(w) \log \left( p(w) \frac{q_\theta(w)}{p(w)} \right) \\ &= -\sum_w p(w) \log p(w) - \sum_w p(w) \log \frac{q_\theta(w)}{p(w)} \\ &= H(p) + \sum_w p(w) \log \frac{p(w)}{q_\theta(w)} \\ &= H(p) + D_{\text{KL}}(p\|q_\theta). \end{aligned}$$

Since  $H(p)$  is constant with respect to  $\theta$ , minimizing  $H(p, q_\theta)$  is equivalent to minimizing  $D_{\text{KL}}(p\|q_\theta)$ .

(d) If  $q_\theta(w_0) = 0$  for some  $w_0$  with  $p(w_0) > 0$ :

- **Mathematical Value:** The term  $p(w_0) \log \frac{p(w_0)}{q_\theta(w_0)}$  in  $D_{\text{KL}}(p\|q_\theta)$  becomes  $p(w_0) \log(\infty) = \infty$ . The term  $-p(w_0) \log q_\theta(w_0)$  in  $H(p, q_\theta)$  becomes  $-p(w_0)(-\infty) = \infty$ . Both  $D_{\text{KL}}(p\|q_\theta)$  and  $H(p, q_\theta)$  become infinite. The identity  $H(p, q_\theta) = H(p) + D_{\text{KL}}(p\|q_\theta)$  still holds as  $\infty = H(p) + \infty$ .
- **Practical Implications:** Optimization breaks down. If an observed sample  $w_i$  (so  $p(w_i) > 0$ ) has  $q_\theta(w_i) = 0$ , the loss  $-\log q_\theta(w_i)$  is infinite. This leads to infinite gradients and numerical instability. Models often use mechanisms like softmax outputs or smoothing to prevent  $q_\theta(w) = 0$ .

(e) **Empirical setting** Given  $n$  i.i.d. samples  $w_{1:n} \sim p$ , their empirical distribution is  $\hat{p}_n(w) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(w_i = w)$ .

$$\text{NLL}(\theta) = -\frac{1}{n} \sum_{i=1}^n \log q_\theta(w_i).$$

The cross-entropy  $H(\hat{p}_n, q_\theta)$  is:

$$\begin{aligned} H(\hat{p}_n, q_\theta) &= -\sum_{w \in W} \hat{p}_n(w) \log q_\theta(w) = -\sum_{w \in W} \left( \frac{1}{n} \sum_{i=1}^n \mathbb{I}(w_i = w) \right) \log q_\theta(w) \\ &= -\frac{1}{n} \sum_{i=1}^n \sum_{w \in W} \mathbb{I}(w_i = w) \log q_\theta(w) = -\frac{1}{n} \sum_{i=1}^n \log q_\theta(w_i) = \text{NLL}(\theta). \end{aligned}$$

Since  $H(\hat{p}_n, q_\theta) = H(\hat{p}_n) + D_{\text{KL}}(\hat{p}_n \| q_\theta)$  and  $H(\hat{p}_n)$  is constant in  $\theta$ , minimizing NLL is equivalent to minimizing  $D_{\text{KL}}(\hat{p}_n \| q_\theta)$ .

Driving  $D_{\text{KL}}(\hat{p}_n \| q_\theta)$  to zero implies  $q_\theta = \hat{p}_n$ . This can lead to *overfitting*:  $q_\theta$  models the specific training sample  $\hat{p}_n$  perfectly, including its noise and deviations from the true  $p$ . Such a model may not generalize well to unseen data from  $p$  because  $\hat{p}_n$  is only an approximation of  $p$ .

(f) **Mutual information forms** For  $(X, Y) \sim p_{XY}$ ,  $I(X; Y) = D_{\text{KL}}(p_{XY} \| p_X p_Y)$ .

1. Derive  $I(X; Y) = H(X) + H(Y) - H(X, Y)$ :

$$\begin{aligned} I(X; Y) &= \sum_{x,y} p_{XY}(x, y) \log \frac{p_{XY}(x, y)}{p_X(x)p_Y(y)} \\ &= \sum_{x,y} p_{XY}(x, y) (\log p_{XY}(x, y) - \log p_X(x) - \log p_Y(y)) \\ &= \sum_{x,y} p_{XY}(x, y) \log p_{XY}(x, y) - \sum_{x,y} p_{XY}(x, y) \log p_X(x) - \sum_{x,y} p_{XY}(x, y) \log p_Y(y) \\ &= -H(X, Y) - \sum_x p_X(x) \log p_X(x) - \sum_y p_Y(y) \log p_Y(y) \\ &= -H(X, Y) + H(X) + H(Y). \end{aligned}$$

2. Interpretation of  $I(X; Y) = 0$ :

$I(X; Y) = 0$  if and only if  $X$  and  $Y$  are statistically independent.

(g) Numerical sanity check Given  $p(0) = 0.1, p(1) = 0.9$  and  $q_\theta(0) = 0.2, q_\theta(1) = 0.8$ . Base-2 logs.  $\log_2(0.1) \approx -3.3219, \log_2(0.9) \approx -0.1520, \log_2(0.2) \approx -2.3219, \log_2(0.8) \approx -0.3219$

1.  $H(p) = -[0.1 \log_2(0.1) + 0.9 \log_2(0.9)] \approx -[0.1(-3.3219) + 0.9(-0.1520)] \approx -[-0.33219 - 0.1368] \approx 0.4690$  bits.
2.  $H(p, q_\theta) = -[0.1 \log_2(0.2) + 0.9 \log_2(0.8)] \approx -[0.1(-2.3219) + 0.9(-0.3219)] \approx -[-0.23219 - 0.28971] \approx 0.5219$  bits.
3.  $D_{\text{KL}}(p \| q_\theta) = p(0) \log_2 \frac{p(0)}{q_\theta(0)} + p(1) \log_2 \frac{p(1)}{q_\theta(1)} = 0.1 \log_2 \frac{0.1}{0.2} + 0.9 \log_2 \frac{0.9}{0.8} = 0.1 \log_2(0.5) + 0.9 \log_2(1.125) \approx 0.1(-1) + 0.9(0.1699) \approx -0.1 + 0.15291 \approx 0.0529$  bits.

Verification:  $H(p) + D_{\text{KL}}(p \| q_\theta) \approx 0.4690 + 0.0529 = 0.5219$ . This matches  $H(p, q_\theta) \approx 0.5219$ .

## Exercise 2: Contrastive Learning

Throughout, fix a **mini-batch size**  $N$  and two encoders

$$f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d, \quad g_\phi : \mathcal{X} \rightarrow \mathbb{R}^d.$$

For each data pair  $(x_i, x'_i)$  we write the embeddings as

$$z_i = f_\theta(x_i), \quad z'_i = g_\phi(x'_i), \quad i = 1, \dots, N.$$

For any  $a, b \in \mathbb{R}^d$  define

$$s(a, b) = \frac{\langle a, b \rangle}{\|a\|_2 \|b\|_2}, \quad \tau > 0 \text{ (temperature)}.$$

We collect the **logits** in the matrix

$$L = [\ell_{ij}]_{i,j=1}^N, \quad \ell_{ij} = \frac{s(z_i, z'_j)}{\tau}.$$

(a) **Soft-max connection**

- (a) **Row-wise soft-max.** For a fixed *anchor row*  $i$  turn the logits  $(\ell_{i1}, \dots, \ell_{iN})$  into the probability distribution

$$p_{\theta}(j | i) = \text{softmax}_j(\ell_{ij}) = \frac{e^{\ell_{ij}}}{\sum_{k=1}^N e^{\ell_{ik}}}.$$

Show that the original InfoNCE objective

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(j = i | i)$$

equals the **cross-entropy**

$$\mathcal{L}_{\text{InfoNCE}} = \frac{1}{N} \sum_{i=1}^N \text{CE}(q(\cdot | i), p_{\theta}(\cdot | i)), \quad q(j | i) = \mathbf{1}\{j = i\}.$$

- (b) **Negatives.** Why are the non-matching pairs  $(z_i, z'_j)$  with  $j \neq i$  called *negatives*?  
Hint: look at the zero entries of  $q(\cdot | i)$ .

(b) **Symmetric loss (CLIP)**

CLIP averages two InfoNCE terms:

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2} [\mathcal{L}_{\text{I} \rightarrow \text{T}} + \mathcal{L}_{\text{T} \rightarrow \text{I}}].$$

- (a) **Image  $\rightarrow$  Text.** Show

$$\mathcal{L}_{\text{I} \rightarrow \text{T}} = \frac{1}{N} \sum_{i=1}^N \text{CE}(q(\cdot | i), p_{\theta}(\cdot | i)),$$

i.e. the cross-entropy over the *rows* of  $L$  (identical to Part A).

- (b) **Text  $\rightarrow$  Image.** Treat each *column*  $j$  as the anchor, define

$$p_{\theta}(i | j) = \frac{\exp(\ell_{ij})}{\sum_{k=1}^N \exp(\ell_{kj})}, \quad q(i | j) = \mathbf{1}\{i = j\},$$

and show the analogous formula for  $\mathcal{L}_{\text{T} \rightarrow \text{I}}$ .

- (c) **All  $2N - 2$  negatives.** Conclude that  $\mathcal{L}_{\text{CLIP}}$  can be written as a *single* loss where both rows and columns of  $L$  are soft-max normalised, so every positive pair  $(i, i)$  is contrasted against all other entries in its row and column.

(c) **Uniformity versus Alignment**

Following Wang & Isola (2020) define

$$\mathcal{A} = \mathbb{E}[\|Z - Z'\|_2^2], \quad \mathcal{U} = \log \mathbb{E}[\exp(-2\|Z - \tilde{Z}\|_2^2)],$$

where  $\tilde{Z}$  is an independent sample from  $p_Z$ .

- (a) Argue qualitatively how minimising  $\mathcal{L}_{\text{InfoNCE}}$  *decreases*  $\mathcal{A}$  (brings paired points together) and *decreases*  $\mathcal{U}$  (spreads all representations).  
(b) Explain why this trade-off becomes harder when the batch contains many modalities or when  $N$  is very large.

## Solution

### (a) Soft-max connection

- (a) **Row-wise soft-max.** The cross-entropy  $\text{CE}(q, p) = -\sum_x q(x) \log p(x)$ . For  $q(j | i) = \mathbf{1}\{j = i\}$ , the sum  $\text{CE}(q(\cdot | i), p_\theta(\cdot | i)) = -\sum_{j=1}^N \mathbf{1}\{j = i\} \log p_\theta(j | i)$  simplifies to  $-\log p_\theta(j = i | i)$  because only the term  $j = i$  is non-zero. Thus,  $\frac{1}{N} \sum_{i=1}^N \text{CE}(q(\cdot | i), p_\theta(\cdot | i)) = -\frac{1}{N} \sum_{i=1}^N \log p_\theta(j = i | i) = \mathcal{L}_{\text{InfoNCE}}$ .
- (b) **Negatives.** Pairs  $(z_i, z'_j)$  with  $j \neq i$  are "negatives" because the target distribution  $q(j | i) = \mathbf{1}\{j = i\}$  assigns them zero probability, indicating they are incorrect matches for the anchor  $z_i$ . The loss pushes their scores  $\ell_{ij}$  down relative to the positive pair's score  $\ell_{ii}$ .

### (b) Symmetric loss (CLIP)

- (a) **Image  $\rightarrow$  Text.** This is  $\mathcal{L}_{\text{I} \rightarrow \text{T}}$ , identical to Part A(a). It uses  $z_i$  (image) as anchor, normalizes logits row-wise for  $p_\theta(j | i)$ , and matches with  $z'_i$  (text).  $\mathcal{L}_{\text{I} \rightarrow \text{T}} = \frac{1}{N} \sum_{i=1}^N \text{CE}(q(\cdot | i), p_\theta(\cdot | i))$ .
- (b) **Text  $\rightarrow$  Image.**  $\mathcal{L}_{\text{T} \rightarrow \text{I}}$  uses  $z'_j$  (text) as anchor and normalizes logits  $\ell_{ij}$  column-wise for  $p_\theta(i | j) = \text{softmax}_i(\ell_{ij})$ . Target is  $q(i | j) = \mathbf{1}\{i = j\}$ . Analogously to A(a),  $\text{CE}(q(\cdot | j), p_\theta(\cdot | j)) = -\log p_\theta(i = j | j)$ . So,  $\mathcal{L}_{\text{T} \rightarrow \text{I}} = \frac{1}{N} \sum_{j=1}^N \text{CE}(q(\cdot | j), p_\theta(\cdot | j))$ .
- (c) **All  $2N - 2$  negatives.**  $\mathcal{L}_{\text{CLIP}} = -\frac{1}{2N} \sum_{i=1}^N (\log p_\theta(j = i | i) + \log p_\theta(k = i | i)_{\text{col}})$ , where  $p_\theta(\cdot)_{\text{col}}$  denotes column-wise softmax. Each positive pair's logit  $\ell_{ii}$  is contrasted with  $N - 1$  row negatives (in  $p_\theta(j = i | i)$ ) and  $N - 1$  column negatives (in  $p_\theta(k = i | i)_{\text{col}}$ ). These are the  $2N - 2$  distinct off-diagonal entries in  $\ell_{ii}$ 's row and column.

### (c) Uniformity versus Alignment

- (a) Minimising  $\mathcal{L}_{\text{InfoNCE}}$  implies maximizing  $s(z_i, z'_i)$  (positive/paired similarity) and minimizing  $s(z_i, z'_k)$  for  $k \neq i$  (negative/unpaired similarity).
- **Alignment ( $\mathcal{A}$  decreases):** Maximizing  $s(Z, Z')$  for paired samples means  $Z, Z'$  become more similar. For unit vectors,  $\|Z - Z'\|_2^2 = 2(1 - s(Z, Z'))$ . So, increasing  $s(Z, Z')$  decreases  $\|Z - Z'\|_2^2$ , thus decreasing  $\mathcal{A}$ .
  - **Uniformity (effect on  $\mathcal{U}$ ):** Minimizing  $s(Z, \tilde{Z})$  for unpaired samples means  $Z, \tilde{Z}$  are pushed apart, increasing  $\|Z - \tilde{Z}\|_2^2$ . For  $\mathcal{U} = \log \mathbb{E}[\exp(-2\|Z - \tilde{Z}\|_2^2)]$ , increased distance  $\|Z - \tilde{Z}\|_2^2$  makes  $\exp(-2\|Z - \tilde{Z}\|_2^2)$  smaller, thus  $\mathcal{U}$  decreases.
- (b) The trade-off becomes harder when:
- **Many Modalities/Semantic Categories:** Distinguishing true pairs from "easy negatives" (e.g., image vs. audio) might be trivial, but the model must also handle "hard negatives" (e.g., image of a cat vs. image of a dog, if the task is instance matching). Fine-grained distinction is needed.
  - **Large  $N$ :** More negatives increase the chance of hard negatives. Keeping the positive score  $s(z_i, z'_i)$  dominant in the softmax denominator becomes more difficult, requiring better separation and potentially leading to instability or overly sharp representations.

**Exercise 3: Variational Auto-Encoders (VAE) and VQ-VAE**

The VAE loss is  $\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \beta D_{\text{KL}}[q(z|x)||p(z)]$ .

**(a) Role of  $\beta$** 

Describe the trade-off controlled by  $\beta$ . What happens if  $\beta = 0$ ? What happens when  $\beta$  is very large?

**(b) Closed-form KL (1-D case)**

Assume a one-dimensional standard normal prior  $p(z) = \mathcal{N}(0, 1)$  and an approximate posterior  $q_\phi(z | x) = \mathcal{N}(\mu(x), \sigma^2(x))$ . Show that

$$D_{\text{KL}}(q_\phi(z | x) || p(z)) = \frac{1}{2}(\mu^2 + \sigma^2 - \log \sigma^2 - 1).$$

*Hint:* write out the KL, collect quadratic and logarithmic terms, and simplify.

**(c) Analogy to VQ-VAE**

In a VQ-VAE the continuous latent vector  $z_e(x) \in \mathbb{R}^d$  produced by the encoder is *quantized* to the nearest codebook vector  $e_k \in \mathbb{R}^d$ :

$$z_q(x) = e_{k^*}, \quad k^* = \arg \min_k \|z_e(x) - e_k\|_2.$$

The loss has three terms

$$\mathcal{L}_{\text{VQ-VAE}} = \mathcal{L}_{\text{rec}} + \underbrace{\|\text{sg}[z_e(x)] - e_{k^*}\|_2^2}_{\text{codebook}} + \beta \underbrace{\|z_e(x) - \text{sg}[e_{k^*}]\|_2^2}_{\text{commitment}},$$

where  $\text{sg}[\cdot]$  denotes the “stop-gradient” operator.

- (i) Explain how the *codebook* term plays a role analogous to the KL divergence in a continuous VAE.
- (ii) Explain the purpose of the *commitment* term and the effect of the hyper-parameter  $\beta$  in this context.

**Solution**

- (a) **Role of  $\beta$**  In the VAE loss  $\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \beta D_{\text{KL}}[q(z|x)||p(z)]$ ,  $\beta$  controls the trade-off between reconstruction quality and the regularization of the latent space.

- $\mathcal{L}_{\text{rec}}$  (reconstruction loss) measures how well the VAE can reconstruct the input  $x$  from the latent representation  $z$ . Minimizing this term encourages the model to learn informative latent codes.
- $D_{\text{KL}}[q(z|x)||p(z)]$  (KL divergence) measures the dissimilarity between the approximate posterior  $q(z|x)$  (encoded distribution for a given  $x$ ) and the prior  $p(z)$  (e.g., a standard normal distribution). Minimizing this term forces  $q(z|x)$  to be close to  $p(z)$ , making the latent space more regular, structured, and enabling meaningful sampling from  $p(z)$  for generation.

The trade-off controlled by  $\beta$  is:

- If  $\beta = 0$ : The KL divergence term is ignored. The VAE becomes a standard autoencoder, focusing solely on minimizing reconstruction error.
- If  $\beta$  is small (but positive): More emphasis is placed on reconstruction quality. The latent space might be less regular, but reconstructions are likely to be better.

- If  $\beta$  is very large: More emphasis is placed on forcing  $q(z|x)$  to match  $p(z)$ . This can lead to a very regular latent space, good for generation. However, it might overly constrain  $q(z|x)$ , forcing it to discard information about  $x$ .

Typically,  $\beta = 1$  is used, corresponding to the true evidence lower bound (ELBO). Varying  $\beta$  (as in  $\beta$ -VAE) can help disentangle factors of variation in the data.

- (b) **Closed-form KL (1-D case)** Given  $p(z) = \mathcal{N}(0, 1)$  and  $q_\phi(z | x) = \mathcal{N}(\mu, \sigma^2)$  (where  $\mu = \mu(x), \sigma^2 = \sigma^2(x)$ ). The KL divergence is  $D_{\text{KL}}(q||p) = \int q(z) \log \frac{q(z)}{p(z)} dz = \mathbb{E}_{q(z)}[\log q(z) - \log p(z)]$ . The log-densities are:  $\log q(z) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(z-\mu)^2}{2\sigma^2}$   $\log p(z) = -\frac{1}{2} \log(2\pi) - \frac{z^2}{2}$ . So,  $\log q(z) - \log p(z) = -\frac{1}{2} \log \sigma^2 - \frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2}$ . Taking the expectation  $\mathbb{E}_{q(z)}[\cdot]$ :

$$\begin{aligned} D_{\text{KL}}(q||p) &= \mathbb{E}_{q(z)} \left[ -\frac{1}{2} \log \sigma^2 - \frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2} \right] \\ &= -\frac{1}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \mathbb{E}_{q(z)}[(z-\mu)^2] + \frac{1}{2} \mathbb{E}_{q(z)}[z^2] \end{aligned}$$

We know  $\mathbb{E}_{q(z)}[(z-\mu)^2] = \text{Var}_{q(z)}(z) = \sigma^2$ . And  $\mathbb{E}_{q(z)}[z^2] = \text{Var}_{q(z)}(z) + (\mathbb{E}_{q(z)}[z])^2 = \sigma^2 + \mu^2$ . Substituting these:

$$\begin{aligned} D_{\text{KL}}(q||p) &= -\frac{1}{2} \log \sigma^2 - \frac{1}{2\sigma^2}(\sigma^2) + \frac{1}{2}(\sigma^2 + \mu^2) \\ &= -\frac{1}{2} \log \sigma^2 - \frac{1}{2} + \frac{1}{2}\sigma^2 + \frac{1}{2}\mu^2 \\ &= \frac{1}{2}(\mu^2 + \sigma^2 - \log \sigma^2 - 1). \end{aligned}$$

(c) **Analogy to VQ-VAE**

- (i) **Role of the codebook term:** The codebook term  $\mathcal{L}_{\text{cb}} = |\text{sg}[z_e(x)] - e_{k^*}|^2$  updates codebook vectors  $e_k$  by pulling the selected vector  $e_{k^*}$  toward the fixed encoder output  $z_e(x)$  (due to stop-gradient). Analogous to the KL divergence term in VAEs which regularizes a continuous latent space by matching encoder outputs to a fixed prior the codebook term structures the discrete latent space by adapting the learnable vectors to match encoder outputs. Both terms ensure useful, structured latent representations for the decoder.

- (ii) **Purpose of the commitment term and  $\beta$ :**

The purpose of commitment term is to ensure that the encoder "commits" to its choice of codebook vector. It forces the continuous output  $z_e(x)$  of the encoder to be close to the chosen discrete codebook vector  $e_{k^*}$ . This prevents  $z_e(x)$  from fluctuating arbitrarily far from the discrete codes, effectively regularizing the encoder's output space to align with the representations available in the codebook. The hyper-parameter  $\beta$  controls the strength of this commitment:

- If  $\beta$  is small: The encoder outputs  $z_e(x)$  are less strongly pulled towards the codebook entries. This might allow  $z_e(x)$  to be more expressive, but could lead to large quantization errors ( $z_e(x)$  far from any  $e_k$ ) if the encoder's outputs are not well aligned with the codebook.
- If  $\beta$  is large: The encoder is heavily penalized if its outputs  $z_e(x)$  are not very close to the chosen  $e_{k^*}$ . This encourages  $z_e(x)$  to map decisively to specific codebook vectors, ensuring the encoder and codebook are "in sync". However, if  $\beta$  is too large, it might overly constrain the encoder, forcing  $z_e(x)$  to collapse to a few codebook vectors or limiting its ability to represent complex information before quantization, potentially harming reconstruction if the codebook isn't yet optimal.

## Exercise 4: Generative Self-Supervision Paradigms

### (a) Autoregressive modelling

**Derive the cross-entropy objective** Show that maximising the autoregressive likelihood

$$\max_{\theta} \prod_{t=1}^N p_{\theta}(x_t | x_{<t})$$

of a discrete token sequence  $x_{1:N}$  is equivalent to minimising the *average token-level cross-entropy loss*

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{t=1}^N \log p_{\theta}(x_t | x_{<t}).$$

**Interpret perplexity** On a held-out set, *perplexity* can be written as

$$\text{PPL} = \exp(\mathcal{L}(\theta)).$$

Interpret the meaning of a perplexity of

- (i) 1,
- (ii) equal to the vocabulary size  $|V|$ , and
- (iii) halfway between those extremes.

### (b) Masked image modeling (MIM)

For a Vision Transformer with patch size  $16 \times 16$ , explain how patch masking affects the spatial receptive field. Design a simple experiment to measure reconstruction quality as a function of the mask ratio. Why do contemporary works often choose  $\rho \approx 0.75$ ?

## Solution

### (a) Autoregressive modelling

**Derive the cross-entropy objective** Maximising the likelihood  $\prod_{t=1}^N p_{\theta}(x_t | x_{<t})$  is equivalent to maximising its logarithm, as the logarithm is a strictly increasing function:

$$\max_{\theta} \log \left( \prod_{t=1}^N p_{\theta}(x_t | x_{<t}) \right) = \max_{\theta} \sum_{t=1}^N \log p_{\theta}(x_t | x_{<t}).$$

Maximising a function is equivalent to minimising its negative:

$$\min_{\theta} - \sum_{t=1}^N \log p_{\theta}(x_t | x_{<t}).$$

Finally, scaling by a positive constant  $1/N$  does not change the optimal parameters  $\theta$ . Thus, this is equivalent to minimising the average token-level cross-entropy loss:

$$\min_{\theta} -\frac{1}{N} \sum_{t=1}^N \log p_{\theta}(x_t | x_{<t}) = \min_{\theta} \mathcal{L}(\theta).$$

**Interpret perplexity** Perplexity is  $\text{PPL} = \exp(\mathcal{L}(\theta))$ . Substituting  $\mathcal{L}(\theta)$ , we get  $\text{PPL} = \exp \left( -\frac{1}{N} \sum_{t=1}^N \log p_{\theta}(x_t | x_{<t}) \right) = \left( \prod_{t=1}^N p_{\theta}(x_t | x_{<t}) \right)^{-1/N}$ . This means PPL is the inverse of the geometric mean of the probabilities assigned by the model to the correct tokens in the sequence. A lower perplexity indicates a better model.



- (i) If  $\text{PPL} = 1$ , then  $\mathcal{L}(\theta) = \log(1) = 0$ . Since each  $p_\theta(x_t | x_{<t}) \in (0, 1]$ , each  $\log p_\theta(x_t | x_{<t}) \leq 0$ . For their sum (and average) to be 0, every term must be 0. This implies  $\log p_\theta(x_t | x_{<t}) = 0$ , so  $p_\theta(x_t | x_{<t}) = 1$  for all  $t = 1, \dots, N$ . **Interpretation:** The model perfectly predicts every token in the sequence with probability 1. It has no uncertainty.
- (ii) If  $\text{PPL} = |V|$ , then  $\mathcal{L}(\theta) = \log |V|$ . This means, on average,  $-\frac{1}{N} \sum_{t=1}^N \log p_\theta(x_t | x_{<t}) = \log |V|$ , or  $\frac{1}{N} \sum_{t=1}^N \log p_\theta(x_t | x_{<t}) = -\log |V| = \log(1/|V|)$ . **Interpretation:** The model is, on average, as uncertain as if it were choosing uniformly from all  $|V|$  tokens in the vocabulary at each step. The average probability assigned to the correct token is  $1/|V|$ .
- (iii) PPL halfway between those extremes: This corresponds to an average log-probability halfway between  $\log(1) = 0$  and  $\log(1/|V|)$ , i.e.,  $\frac{1}{2} \log(1/|V|) = \log(1/\sqrt{|V|})$ . If  $\text{PPL} = \sqrt{|V|}$ , then  $\mathcal{L}(\theta) = \log \sqrt{|V|}$ . **Interpretation:** The model is, on average, as uncertain as if it were choosing uniformly among  $\sqrt{|V|}$  options. The geometric mean probability assigned to the correct token is  $1/\sqrt{|V|}$ .

(b) **Masked image modeling (MIM)**

1. **Setup:** Use a standard ViT architecture (e.g., ViT-Base) and an image dataset
2. **Variable:** Define a set of mask ratios  $\rho$  to test, e.g.,  $\rho \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$ .
3. **Training:** For each  $\rho$ :
  - Pre-train the ViT using MIM: randomly mask  $\rho N_p$  patches (where  $N_p$  is total patches) and train the model to reconstruct the original masked patches.
  - Use consistent training hyperparameters (epochs, learning rate, etc.) for all  $\rho$ .
4. **Evaluation:** For each trained model:
  - On a held-out validation set, apply the same masking ratio  $\rho$  used during its training.
  - Measure reconstruction quality using a metric like Mean Squared Error (MSE) or PSNR between original and reconstructed masked patches.
5. **Analysis:** Plot the reconstruction quality (e.g., MSE on y-axis) against the mask ratio  $\rho$  (on x-axis) to observe the trend.

**Why do contemporary works often choose  $\rho \approx 0.75$ ?**

- **Challenging pretext task:** A high mask ratio (like 0.75) makes the reconstruction task very difficult. This forces the model to learn rich, non-trivial representations by understanding long-range dependencies and holistic scene context from sparse visible information. This often leads to better generalization on downstream tasks.
- **Computational efficiency:** In ViTs, self-attention complexity is quadratic in the number of input tokens. By masking 75% of patches, only 25% are processed by the main encoder. This drastically reduces computational cost and memory usage during pre-training, enabling larger models or faster training.
- **Empirical success:** Seminal works like Masked Autoencoders (MAE) demonstrated that high mask ratios like 0.75 can lead to state-of-the-art performance on various downstream vision tasks. This suggests natural images possess high redundancy, and powerful features can be learned even from very sparse inputs.

**Exercise 5: Multimodal Alignment - Understanding****(a) Positives and negatives in CLIP**

For a batch of  $N$  image-text pairs  $(I_i, T_i)$ , how many positive and how many negative pairs are considered for a *single* image  $I_i$  when computing the image-to-text loss component?

**(b) Extending CLIP to three modalities**

Suppose you add an audio encoder producing embeddings  $A_i$ . Formulate a contrastive loss that aligns all three modalities (I, T, A). How many distinct pairwise loss terms are required in total? Compare your idea to OmniBind from the lecture.

**(c) Sigmoid loss (SigLIP)**

Unlike CLIP, SigLIP replaces the softmax with a pairwise sigmoid loss. Write the SigLIP loss for one positive pair  $(I_i, T_i)$  and explain why negatives are sampled differently compared to CLIP. What happens to the influence of easy vs hard negatives on the image and text encoders' updates?

**(d) JEPA versus CLIP**

Briefly compare the objective of a Joint Embedding Predictive Architecture (JEPA) with CLIP. Why is JEPA considered predictive rather than contrastive? List two key advantages JEPA claims over contrastive approaches, and briefly explain how its architecture enables each advantage.

**(e) JEPA Masking** Compare the effects of using a single large contiguous mask versus many small scattered masks on the learned representation. How might the optimal mask strategy differ between images and audio waveforms?**(f) MultiMAE**

MultiMAE masks a random subset of modalities and reconstructs the missing ones. Formulate the training objective for a three-modal setup (I, T, A) and discuss how it differs from both CLIP and JEPA.

**(g) MultiMAE Loss** MultiMAE linearly combines per-modality reconstruction losses. What could happen if one modality's loss dominates the total objective?**(h) MultiMAE Shared encoder vs. modality-specific decoders**

Give two reasons why the authors attach small, separate decoders instead of one large shared decoder. What possible downside might arise when decoders are too lightweight or too isolated from each other?

**Solution****(a) Positives and negatives in CLIP** For a single image  $I_i$  in a batch of  $N$  image-text pairs  $(I_k, T_k)$ :

- **Positive pairs:** 1 (the pair  $(I_i, T_i)$ ).
- **Negative pairs:**  $N - 1$  (the pairs  $(I_i, T_j)$  for all  $j \neq i$ ).

**(b) Extending CLIP to three modalities** For modalities I, T, A (Image, Text, Audio), we need to align each pair. There are 3 unordered pairs: (I,T), (I,A), (T,A). For each pair (X,Y), CLIP uses two directed losses:  $\mathcal{L}_{X \rightarrow Y}$  and  $\mathcal{L}_{Y \rightarrow X}$ . Thus,  $3 \times 2 = 6$  distinct pairwise InfoNCE loss terms are required in total.

The total loss would average these 6 terms.

- (c) For a positive pair  $(I_i, T_i)$  with similarity  $s_{ii}$ , and negative texts  $T_j$  with similarities  $s_{ij}$ , the SigLIP loss for  $I_i$  is:  $L_i = -\log(\text{sigmoid}(s_{ii})) - \sum_{j \neq i} \log(1 - \text{sigmoid}(s_{ij}))$  (This encourages  $s_{ii}$  to be high and  $s_{ij}$  to be low).

Why negatives are sampled differently: CLIP uses all other in-batch items as negatives (due to the softmax denominator). SigLIP's pairwise loss decouples negatives. This means SigLIP can use a different number of negatives, potentially sampled from a larger pool outside the current batch, because there's no single normalizing sum over all batch items.

Influence of easy vs. hard negatives:

- **CLIP (softmax):** Hard negatives (high similarity to anchor) dominate the loss and gradients. Easy negatives have little impact.
- **SigLIP (sigmoid):** Easy negatives (very low similarity) also contribute little. Hard negatives provide a strong learning signal. SigLIP is less dominated by a few extremely hard negatives compared to softmax, allowing a broader range of hard negatives to influence learning more smoothly.

(d) **JEPA versus CLIP**

- **CLIP:** Learns by contrasting similar image-text pairs against dissimilar ones, pushing their embeddings apart in a shared space.
- **JEPA:** Learns by predicting the embedding of a masked part of an input (e.g., image patch) from the embedding of an unmasked part, often within the same modality.

Why JEPA is predictive: Its main goal is to predict representations (embeddings) of hidden content, using a regression-like loss in embedding space, not just to distinguish between positive and negative examples.

Two key advantages JEPA claims:

1. **Avoids issues with negative sampling/feature suppression:** Contrastive methods can sometimes suppress useful features if they appear in "false negatives." JEPA doesn't need explicit negatives.  
**How architecture enables this:** It predicts stable target embeddings (from an EMA target encoder), encouraging the model to learn comprehensive features from the context rather than just discriminative ones.
2. **Learns more semantic/abstract representations:** Predicting embeddings (abstract data) rather than raw pixels avoids focusing too much on low-level details.  
**How architecture enables this:** The loss is on embeddings. This forces the model to capture information relevant at that level of abstraction, not just pixel-perfect reconstruction.

(e) **JEPA Masking**

- **Single large contiguous mask:** Forces learning of global, long-range dependencies and holistic understanding to infer the large missing region. May yield more abstract features.
- **Many small scattered masks:** Relies more on local context, textures, and fine-grained details for "inpainting" small gaps.
- **Optimal strategy for images vs. audio:**
  - **Images:** A mix can be good. Large masks for objects/layout; small masks for texture.

- **Audio Waveforms:** Large contiguous masks (masking a time segment) are often very effective. This helps learn temporal structure (like rhythm, melody, or predicting what sound comes next). Small scattered masks might be less natural for capturing the flow of audio unless applied to a spectrogram.
- (f) **MultiMAE** If modality  $k$  (e.g., image) is masked, the model uses unmasked modalities (e.g., text, audio) to predict the original content of  $k$ . The loss is typically a sum of reconstruction losses for the masked modalities:  $L_{MultiMAE} = \sum_{k \in \{\text{masked modalities}\}} \text{weight}_k \cdot \text{ReconstructionLoss}_k(\text{predicted } x_k, \text{original } x_k)$  (e.g., MSE for images, cross-entropy for text tokens).

Differences:

- **vs. CLIP:** MultiMAE reconstructs raw data (generative); CLIP contrasts existing pairs (discriminative).
  - **vs. JEPA:** MultiMAE reconstructs raw data (pixels, tokens); JEPA predicts abstract embeddings.
- (g) **MultiMAE Loss** If one modality's reconstruction loss (e.g.,  $\mathcal{L}_{\text{rec},I}$ ) dominates  $\mathcal{L}_{\text{MultiMAE}}$ : This means the model might:
- Focus too much on learning features to reconstruct that one modality.
  - Neglect learning for other modalities or their cross-modal relationships.
  - Result in biased representations and poor performance on tasks involving the less dominant modalities.
- (h) **MultiMAE Shared encoder vs. modality-specific decoders** Reasons for small, separate decoders:
1. **Handle Different Output Types:** Images (pixels), text (tokens), and audio (waveforms) have very different structures. Specialized decoders are more efficient and effective for each. The bulk of parameters stays in the shared encoder.
  2. **Modularity & Scalability:** It's easier to add a new modality by just designing its specific encoder and small decoder, without overhauling a single large decoder.

Downside of too lightweight/isolated decoders:

1. **Bottleneck:** If a decoder is too simple, it might not be able to reconstruct the modality well, even if the shared encoder provides good information. This weakens the learning signal.
2. **Limited Fine-tuning:** Completely isolated decoders can't use information from other modalities at the very last stage of reconstruction for potential refinements.